

Research

Repeatability, reproducibility, and observer variability of body composition assessment with single-slice lumbar MRI

Magdalena Nowak¹ · Luis Núñez¹ · Charles Edward Hill¹ · Tom Davis¹ · Luis Felipe Cardiel¹ · Gary Robert Schooler² · Asimenia Mermekli³ · Roberto Salvati¹ · Helena Thomaides Brears¹ · Michele Pansini^{4,5} · Matthew David Robson¹

Received: 14 October 2024 / Accepted: 28 January 2025

Published online: 13 February 2025

© The Author(s) 2025 **OPEN**

Abstract

Purpose The single-slice MRI at the third lumbar vertebra (L3) offers an efficient way to assess body composition but evidence on the reliability of this method is scarce. This study evaluates the accuracy and precision of the single-slice L3 method for adipose and muscle tissue measurements.

Methods The technical performance of single-slice (L3) MRI body composition measurements was assessed in a prospective, single-center study of 12 participants, focusing on scan-rescan repeatability, cross-scanner reproducibility, and analyst variability. Participants were scanned using Philips 1.5 T (reference scanner) and Philips 3 T scanners. Additionally, retrospective data from multiple imaging centers from 36 participants were analyzed to evaluate inter-device and inter-observer (analyst vs. radiologist) variability across a wide range of body types, scanner manufacturers (Siemens, Philips, GE) and field strengths (1.5 T and 3 T). Blinded analyses were performed for visceral adipose tissue (VAT) and subcutaneous adipose tissue (SAT) cross-sectional areas (CSA), VAT and SAT indices, VAT/SAT ratio, skeletal muscle CSA, skeletal muscle index (SMI), psoas muscle CSA, and psoas muscle index.

Results Single-slice MRI-based body composition measurements showed high scan-rescan repeatability (CoV: 1.5%–7.9%, ICC: 0.97–1.0), with low repeatability coefficients (RC) across key metrics, including 12 cm² for SAT CSA, 15 cm² for VAT CSA, 5 cm² for skeletal muscle CSA, and 1.4 cm² for psoas muscle CSA. Cross-scanner reproducibility was consistent (CoV: 2.3%–15%, ICC: 0.90–1.0). Inter- and intra-analyst variability was minimal (CoV: 0.5%–5.0%, ICC: 0.98–1.0). Analyst-radiologist comparisons showed near-perfect correlations ($r = 0.97$ – 1.00 , $p < 0.001$) and excellent reliability (ICC: 0.96–1.0).

Conclusion The results demonstrate that the MRI-based single-slice method at the L3 vertebral level provides accurate, repeatable, and reproducible measurements of adipose and muscle tissue across diverse body types, consistent between trained analysts and experienced radiologists. This study establishes technical validity of the single-slice L3 method and supports its potential for use in clinical and research settings.

Keywords Single-slice body composition · Magnetic resonance imaging · Repeatability · Reproducibility · L3

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s44352-025-00005-2>.

✉ Michele Pansini, michele.pansini@eoc.ch | ¹Perspectum Ltd, Gemini One, 5520 John Smith Drive, Oxford OX4 2LL, UK. ²Cincinnati Children's Hospital Medical Center, University of Cincinnati College of Medicine, Cincinnati, OH, USA. ³University of Oxford, John Radcliffe Hospital, Oxford OX3 9DU, UK. ⁴Department of Radiology, Oxford University Hospitals NHS Foundation Trust, Oxford, UK. ⁵Istituto Di Imaging Della Svizzera Italiana (IIMS), Clinica Di Radiologia EOC, Lugano, Switzerland.



1 Introduction

Obesity and sarcopenia are significant public health issues with a rising prevalence globally, both contributing to increased healthcare costs and reduced quality of life [1–4]. Obesity increases the risk of chronic diseases such as cardiovascular disease [5], chronic kidney disease [6, 7], diabetes [8], metabolic dysfunction-associated liver disease (MASLD) [9, 10], and certain types of cancer [11, 12]. Similarly, sarcopenia, defined as the loss of skeletal muscle mass and strength, is associated with increased adverse outcomes including falls, functional decline, frailty, diabetes, and mortality [13–16]. Anthropometric measures such as body mass index (BMI), waist circumference, and waist-to-hip ratio are widely used to estimate fat distribution but they are inadequate for accurately predicting individual fat distribution and metabolic risk [17, 18]. BMI, in particular, is a crude measure that fails to differentiate between fat and muscle mass [18, 19], both of which are crucial for assessing metabolic health. This limitation is particularly evident in conditions like sarcopenia and cachexia, where significant muscle wasting occurs. In sarcopenia, age-related muscle loss may be accompanied by increased adiposity, which can result in a ‘normal’ BMI despite significant muscle loss. Cachexia, associated with chronic illness, involves rapid muscle and weight loss that may be reflected in a reduced BMI, which fails to fully capture the underlying changes in body composition. Sarcopenia is associated with a higher rate of mortality (pooled odds ratio [OR] of 3.596, 95% confidence interval [CI]: 2.96–4.37) [20], functional decline (pooled OR of 3.03, 95% CI: 1.80–5.12) [21, 22], a higher risk of hospitalization (pooled hazards ratio [HR] of 1.57, 95% CI: 1.26–1.94), and lower quality of life [21]. Therefore, there is a critical need for advanced body composition analysis tools that can provide an accurate adipose and muscle tissue assessment. Such tools are essential not only for the effective evaluation and management of conditions such as obesity and sarcopenia but also hold significant value in clinical trials for monitoring the efficacy of various weight loss interventions, including pharmacological agents [23], bariatric surgery [23, 24], and lifestyle modifications [23, 25]. Furthermore, these tools are useful in predicting clinical outcomes beyond mere weight loss measures. They can provide important information about a patient’s potential response to major interventions such as surgery [26] or chemotherapy [27], allowing for more personalized and effective treatment planning. This predictive capability enhances risk assessment and can guide clinical decision-making, potentially improving patient outcomes across various medical disciplines [28].

Whole-body magnetic resonance imaging (MRI) is the gold standard for assessing the skeletal muscle and adipose tissue volumes [29]. It has been validated to reliably and accurately measure body compartments [30–35], and its lack of ionizing radiation, combined with its applicability to nearly all cohorts and ages, makes it suitable for repeated scans in longitudinal studies and for monitoring interventions [29]. However, its widespread application is limited by high costs, time-consuming manual post-processing, a lack of reference ranges, and the increasing demands on an already overburdened radiology workforce, further hampering its widespread use. Computed tomography (CT), while more cost-effective, utilizes ionizing radiation, which limits its use, particularly for repeated scans and in vulnerable populations [36, 37]. Alternatives to whole-body solutions have focused on measurements from single- or few-slice segmentations. Previous research has shown that positioning single-slice measurements at the level of third lumbar vertebra (L3) is optimal in terms of reproducibility and correlations with whole-body compartments [38, 39], thereby simplifying data acquisition, reducing assessment time, complexity and overall costs, as required for diagnostic applications in clinical settings. In recent years, AI-based methods have been developed to automate body composition analysis [40–42], demonstrating significant potential for rapid data evaluation. However, the integration of these technologies into clinical practice is uncertain due to the necessary caution among users, the challenges of using cloud computing solutions for managing patient sensitive data, and the scrutiny from regulatory bodies. Despite the advancements in AI, there is still limited foundational knowledge on the technical performance of single-slice measurements of adipose and muscle tissue using manual segmentation techniques. Comprehensive technical validation is essential not only for broader adoption of this methodology in routine clinical workflows and clinical trials but also for establishing a reliable benchmark against which emerging technologies can be evaluated. This study aims to assess the accuracy and precision of MRI-based single-slice L3 measurements of adipose and muscle tissue.

2 Materials and methods

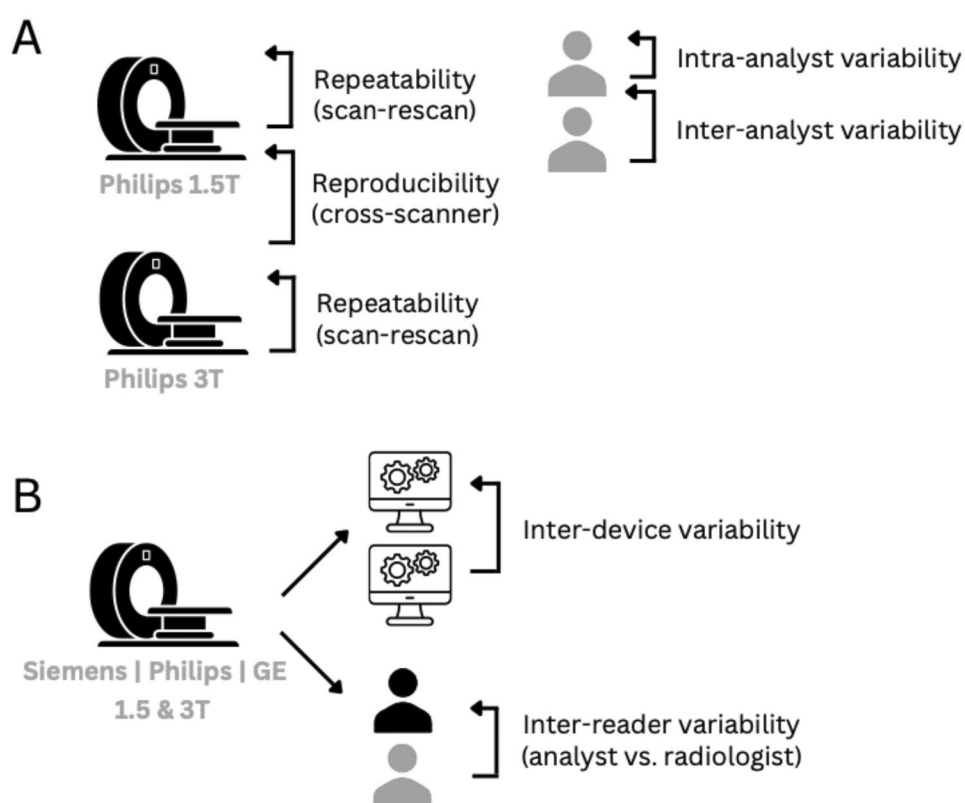
2.1 Study design and population

The technical performance of single-slice body composition measurements was evaluated in a total of 48 individuals, including 12 from a prospective cohort and 36 from a retrospective cohort. All studies received appropriate approvals from the West of Scotland Research Ethics Committee (REC reference: 21/WS/0066) and were performed in accordance with the principles of the Declaration of Helsinki. All participants provided informed consent to participate, including those who had previously taken part in ethically approved studies and consented to the use of their data.

Two primary software tools were used in this study for body composition analysis. The first tool, OsiriX MD (Pixmeo, Switzerland), is an FDA-cleared, commercially available medical imaging software used as the reference standard in this study. It provides general-purpose tools for medical image analysis, including both manual and semi-automated segmentation capabilities. This will hereafter be referred to as the "reference software." The second tool, VitruvianScan™ (Perspectum Ltd., UK), is a specialized, FDA-cleared, single-slice (2D) body composition MRI analysis software. It incorporates manual segmentation tools specifically designed for adipose and muscle tissue quantification at the L3 vertebral level. This will hereafter be referred to as the "study software." Both software packages were used to analyze the same MRI datasets, allowing for comparison of their performance in body composition assessment.

In the prospective single-center study, 12 participants were recruited. A prospective study allowed careful evaluation of the repeatability and reproducibility of the methods. Participants were scanned on Philips 1.5 T (reference scanner) and Philips 3 T scanners (Fig. 1A). For scan-rescan repeatability, each participant was scanned twice under the same conditions on the same day, with the patient leaving the MRI table between scans and being re-localized and re-registered with a new blinding code for each scan, with a maximum 30-min interval between the scans. Cross-scanner reproducibility involved scanning each participant on the same day with both scanners, with a maximum 1-h interval between the scans (Fig. 1A). Inter- and intra-analyst variability assessments were conducted on data acquired from the Philips 1.5 T (reference) scanner. Adult subjects aged 18 years or older, with BMI between 20 and 35 kg/m²,

Fig. 1 Study design. **A** Prospective testing: repeatability, reproducibility, and intra/inter-analyst variability; **B** Retrospective testing: inter-device and inter-reader variability (analysts vs. radiologists)



and with a range of body types (from muscular to frail) were included. Exclusion criteria comprised the presence of any MRI contraindications (e.g. pacemaker, shrapnel injury, severe claustrophobia) [43].

In addition, retrospective data from multiple imaging centers from 36 participants were selected to test the methods across a wide range of body types, scanner manufacturers (Siemens, Philips, GE), and field strengths (1.5 T and 3 T) (Fig. 1B). To achieve balanced representation, six participants were chosen for each combination of scanner manufacturer and field strength, with equal distribution across sex (3 females, 3 males) and body types (lean, normal weight, obese). Body type classification was based on appearance, with "lean" individuals showing visibly lower fat and higher muscle mass, "normal weight" individuals having average fat and muscle appearance, and "obese" individuals having higher visible fat levels. Inclusion criteria ensured a diverse dataset, covering both sexes, a range of body types, BMI values between 20 and 35 kg/m², different scanner manufacturers and strengths (Siemens 1.5 T, Siemens 3 T, Philips 1.5 T, Philips 3 T, GE 1.5 T and GE 3 T), and known height for all participants. This dataset was used for inter-device assessments, comparing the performance of the study software with the reference software. Additionally, inter-reader assessment was performed by having trained analysts and radiologists independently analyze the same images using the study software to compare performance across different reader groups (Fig. 1B).

2.2 Data acquisition

In the prospective testing, all individuals were scanned using Philips Ingenia 1.5 T and Philips dStream 3 T. Data acquisitions were performed at the Leeds General Infirmary (The Leeds Teaching Hospitals NHS Trust), United Kingdom.

The prospective MRI data were acquired using a 3D SPGR (Spoiled Gradient Echo) sequence with the Dixon option (2 echo times) in the axial plane, a flip angle of 10° (1.5 T) or 6° (3 T), and minimum repetition time (TR of 6.7 ms at 1.5 T, TR of 4.6 ms at 3 T). The maximum field of view (FOV) was set in both the right-left (RL) and anterior–posterior (AP) directions. Imaging was performed with an in-plane resolution of 2 × 2 mm² and a slice thickness of 6 mm, with a reconstructed slice thickness set to 3 mm (interpolation on). Parallel imaging techniques were utilized with acceleration factors of 2 in the phase-encoding direction and 2 in the slice-encoding direction with SENSE. A single average was used for acquisitions at 1.5 T, while two averages were used for acquisitions at 3 T. This decision was made because the TR at 3.0 T is shorter, and our objective was to limit the number of slices, thereby keeping the acquisition time to approximately 12 s. Data were collected in a single breath-hold and centered on the L3 vertebra. The detailed MRI acquisition parameters are described in Supplementary Table S1.

The retrospective MRI data had been acquired using a 3D SPGR Dixon sequence. The in-plane resolution was between 1 × 1 mm² and 4 × 4 mm². The acquired slice thickness was between 1 and 10 mm. The flip angle was required to be less than or equal to 16° and 12° at 1.5 T and 3 T, respectively.

2.3 Data processing

Anonymised MR data were analyzed using both the study software and the reference software, as described earlier. Throughout all analyses, data analysts (hereafter referred to as 'analysts') and radiologists remained blinded to the subject identities, scanner manufacturers/models, field strengths, scan session details, and any analyses or results produced by other radiologists or analysts. The analyses were performed by trained analysts with a minimum of 2 years of experience in MRI quantitative analysis and/or radiologists with a minimum of 5 years of experience in abdominal MRI. Analysts and radiologists underwent comprehensive training on the study software and had prior experience performing single-slice analyses at the L3 vertebral level.

Body composition analysis involved manual segmentations on a single L3 slice to obtain cross-sectional area (CSA) measurements (in cm²) for four classes; the visceral adipose tissue (VAT), the subcutaneous adipose tissue (SAT), the skeletal and the psoas muscles (Fig. 2). Manual segmentation was selected for this study to maintain high-quality control over the measurements.

The assessment of observer variability included: prospective intra- and inter-analyst evaluations as well as a retrospective inter-reader comparison (analysts vs. radiologists). For the prospective intra-analyst variability assessment, the same analyst analyzed a sub-sample of scans from the reference scanner twice (analysis 1 and analysis 2), presented in a random order and under blinded conditions, with a maximum of 4 days between the reads. Three analysts processed the same set of cases twice, with the anonymized cases presented randomly each time. For inter-analyst assessment, three analysts analyzed the same sub-sample of scans from the reference scanner under blinded conditions. For the retrospective inter-reader variability assessment (analysts vs. radiologists), a "gold standard" measurement was created

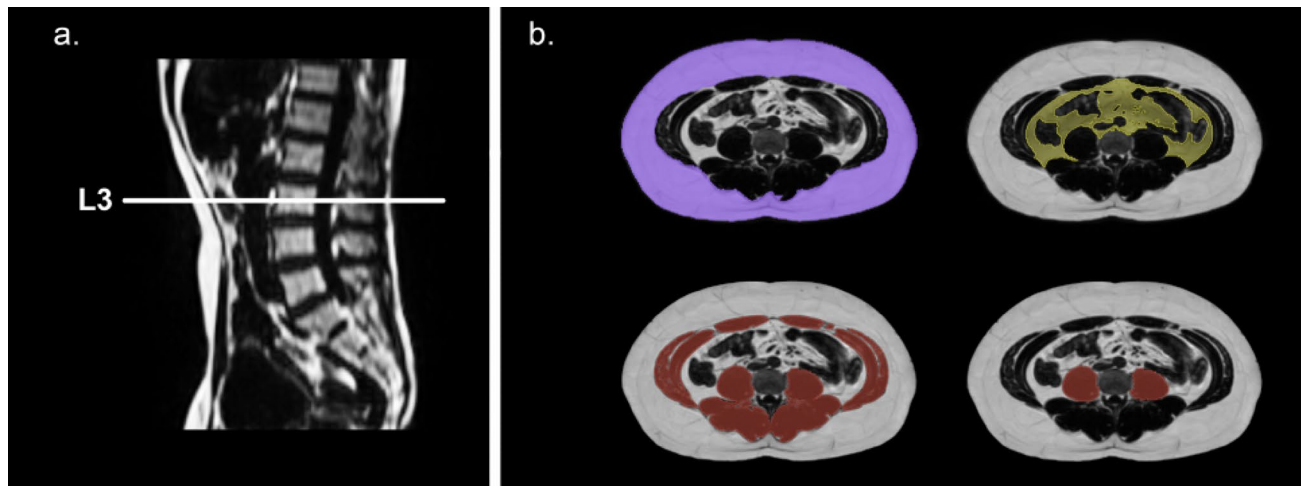


Fig. 2 **a** Axial slice position at the level of L3 vertebra; **b** cross-sectional colored masks for subcutaneous adipose tissue (SAT, top left), visceral adipose tissue (VAT, top right), skeletal muscle (bottom left), and psoas muscle (bottom right) all superimposed on an image of the fat

on the median of successful measurements by the three radiologists using the study software. This “gold standard” was then used to evaluate the performance of each of the analysts. The median values of measurements were used to create a consistent reference point for comparing the analysts’ performance. This approach provided a reliable, consensus-based benchmark, reducing potential biases associated with relying on a single radiologist’s measurement.

The retrospective inter-device assessment compared results from the study software and the reference software on the same set of images. Three radiologists used the study software, while one of these radiologists (designated as the reference radiologist) used the reference software. The median results from the three radiologists using the study software were compared with the results from the reference radiologist using the reference software. To generate results using the reference software, the reference radiologist was presented with a single axial slice at the center of the L3 vertebrae. Initially, a region-growing tool within the software, utilizing image-dependent threshold values, was used to generate coarse masks for the four tissue classes. Subsequently, the reference radiologist edited these masks using a brush tool, where necessary, to generate the final output masks. The regions of interest (ROIs) were then exported, and the CSA measurements parsed from the resulting output file.

Within the study software, pre-processing was performed to generate a signal fat fraction (sFF) map calculated on a voxel-wise basis ($F/(F + W)$), where F represents the signal in the fat image generated from the 3D SPGR Dixon acquisition, and W represents the signal in the corresponding water image [44]. Analysts were presented with sagittal views of the fat, water, and calculated sFF maps, with a line indicating the central slice (Fig. 2a). This central slice was identical to the slice used by the radiologist in the reference device. Upon loading the central slice, the analysts used a circular brush tool to segment regions corresponding to a specific tissue class. During this interactive process, as brush strokes were applied to the image to generate an output segmentation mask, voxels within the bounds of the brush were only added to the segmentation mask if specific criteria were met in the intensity values of the corresponding voxels in the sFF image (Fig. 3). For VAT and SAT classes, voxels were included where $sFF > 0.5$. For skeletal muscle and psoas classes, voxels were included where $sFF < 0.5$.

After analyses were completed by analysts and radiologists and measurements exported, these data were subsequently evaluated using an automated statistical pipeline.

2.4 Statistical analysis

For the prospective testing of repeatability, reproducibility, and intra/inter-analyst variability, the following statistical analyses were performed: Bland Altman analysis (bias and 95% limits of agreement, LoA) [45, 46], standard deviation (SD) of differences, mean coefficient of variation (CoV, in %), intra-class coefficient (ICC, two-way mixed effects, absolute agreement, single measurement), and repeatability coefficient (RC) or reproducibility coefficient (RDC). The RC/RDC, an estimate of the maximum difference that only 5% of pairs of measurements on the same subject will exceed asymptotically, was calculated according to standard formulae (approximately 2.77 times the within-subject standard deviation) [47, 48]. ICC is a reliability index that reflects both degree of correlation and agreement between measurements, whereby

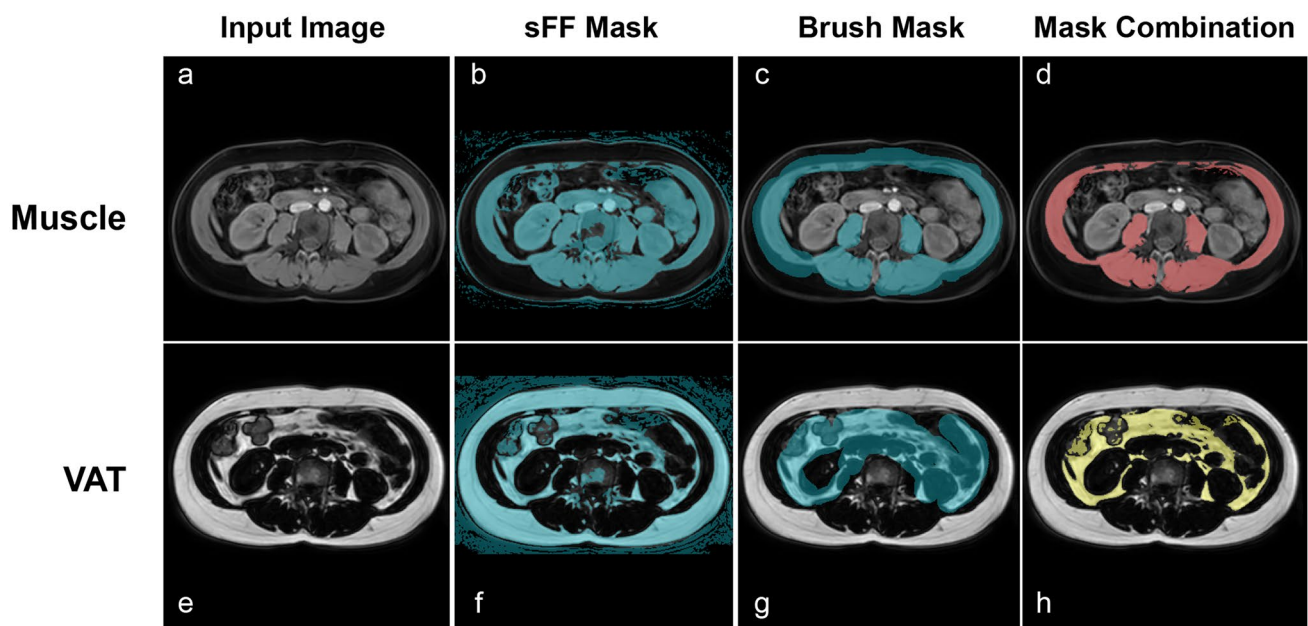


Fig. 3 Segmentation process for muscle and adipose tissue using the study software. The figure illustrates the steps in generating segmentation masks for muscle and fat tissues. **a, e** The original input images; **b, f** The signal fat fraction (sFF) masks, where thresholds < 0.5 and > 0.5 are used to segment muscle and adipose tissue, respectively; **c, g** Brush stroke applied by a user to an area identified as a specific class; **d, h** Final mask is generated by combining the sFF and brush masks, showing the intersection of the two

agreement was classified as follows: ICC > 0.9 , excellent; $0.9-0.75$, good; $0.75-0.5$, moderate; < 0.5 , poor [49]. These statistical analyses were also performed for the retrospective evaluation of intra- and inter-reader (analyst vs. radiologist) variability. In addition, Pearson correlation coefficients and corresponding p-values were calculated for inter-reader assessments. The statistical analyses were performed for all CSA (cm^2): VAT, SAT, skeletal muscle, and psoas muscle; VAT/SAT ratio (no units); and VAT, SAT, skeletal muscle, and psoas indices (cm^2/m^2). Inter-device variability assessments were conducted only for CSAs. The indices were calculated by indexing the body composition CSA measurements (cm^2) to height by dividing it by the height squared (m^2). These height-standardized measurements can be used for comparison between individuals of different height.

In the analysis of participant demographics, values are presented as mean $\pm$ standard deviation (SD) for normally distributed data and as median with interquartile range (IQR) for non-normally distributed data. Descriptive statistics were also calculated for all prospective body composition measurements with all study subjects pooled.

The required sample size for the performance testing of the MRI body composition measurements was estimated based on Bland Altman estimates using previously described methodology [50].

All statistical analyses were performed using scripts written in Python (v3.10).

3 Results

3.1 Participant characteristics

In the prospective data analysis, 12 participants were included (5 females), with a mean age of $31 (\pm 10)$ years and a mean BMI of $23.6 (\pm 4.3) \text{ kg/m}^2$. Additionally, 36 individuals (18 females) were included in the retrospective inter-device and inter-reader analysis, with a median age of 31 (24–43) years and a median BMI of $24.9 (21.5-28.0) \text{ kg/m}^2$. Individual subject characteristics are reported in Supplementary Tables S2 and S3.

Table 1 Descriptive statistics of the prospective study cohort

Measurement	Median	Q1	Q3
Psoas muscle CSA (cm ²)	23.5	16.4	34.3
Psoas muscle Index (cm ² /m ²)	7.7	5.7	10.4
Skeletal muscle CSA (cm ²)	125.5	106.0	174.5
Skeletal muscle Index (cm ² /m ²)	41.1	38.0	50.8
SAT CSA (cm ²)	124.0	101.0	178.5
SAT Index (cm ² /m ²)	44.1	30.7	61.0
VAT CSA (cm ²)	65.9	47.4	93.6
VAT Index (cm ² /m ²)	25.4	14.6	38.3
VAT/SAT Ratio	0.49	0.29	0.80

CSA cross-sectional area, SAT subcutaneous adipose tissue, VAT visceral adipose tissue

3.2 Median values for single-slice body composition measurements

Descriptive statistics (median and quartiles of each measurement variable) for the prospective study cohort (N = 12) are presented in Table 1.

3.3 Scan-rescan repeatability

The scan-rescan repeatability results (bias, LoA, SD, ICC, RC, and CoV) for each body composition measurement for the reference scanner (Philips 1.5 T) are reported in Table 2. The RC values were: 1.4 cm² for psoas muscle CSA, 0.44 cm²/m² for psoas muscle index, 5.1 cm² for skeletal muscle CSA, 1.9 cm²/m² for skeletal muscle index, 12 cm² for SAT CSA, 3.5 cm²/m² for SAT index, 15 cm² for VAT CSA, 4.5 cm²/m² for VAT index, and 0.099 for VAT/SAT ratio. All measurements showed relatively low variability (CoV 1.5% to 7.9%) and excellent reliability (ICC 0.98 to 1.0) across the tested scanner field strengths (Table 2 and Supplementary Table S4). Overall, VAT-related measurements consistently showed greater variability compared to measurements across all testing conditions (CoV 5.0% to 7.9%).

3.4 Cross-scanner reproducibility

The reproducibility results (bias, LoA, SD, ICC, RDC, and CoV) for the cross-scanner comparisons between the Philips 1.5 T and Philips 3 T are reported in Table 3. The RDC values were: 4.6 cm² for psoas muscle CSA, 1.6 cm²/m² for psoas muscle index, 7.5 cm² for skeletal muscle CSA, 2.8 cm²/m² for skeletal muscle index, 26 cm² for SAT CSA, 8.4 cm²/m² for SAT index, 42 cm² for VAT CSA, 12 cm²/m² for VAT index, and 0.27 for VAT/SAT ratio. All measurements showed low to moderate variability (CoV 2.3% to 15%) and excellent reliability (ICC 0.90 to 1.0).

Table 2 Summary of scan-rescan repeatability results for single-slice body composition measurements for the reference scanner (Philips 1.5 T)

Measurement	Bias	Lower LoA	Upper LoA	SD	ICC	RC	CoV (%)
Psoas muscle CSA (cm ²)	− 0.44	− 1.5	0.65	0.56	1.0	1.4	2.0
Psoas muscle Index (cm ² /m ²)	− 0.15	− 0.49	0.19	0.17	1.0	0.44	2.0
Skeletal muscle CSA (cm ²)	0.12	− 5.2	5.4	2.7	1.0	5.1	1.5
Skeletal muscle Index (cm ² /m ²)	0.050	− 2.0	2.1	1.0	0.99	1.9	1.7
SAT CSA (cm ²)	− 1.8	− 13	9.7	5.9	1.0	12	1.9
SAT Index (cm ² /m ²)	− 0.52	− 4.0	3.0	1.8	1.0	3.5	1.9
VAT CSA (cm ²)	1.7	− 14	17	7.9	0.99	15	5.0
VAT Index (cm ² /m ²)	0.38	− 4.3	5.0	2.4	0.98	4.5	5.0
VAT/SAT Ratio	0.012	− 0.089	0.11	0.051	0.98	0.099	5.9

CoV coefficient of variation (%), CSA cross-sectional area, ICC intraclass coefficient, LoA limits of agreement, RC repeatability coefficient, SAT subcutaneous adipose tissue, SD standard deviation, VAT visceral adipose tissue

Table 3 Summary of cross-scanner reproducibility results for single-slice body composition measurements (Philips 3 T vs. Philips 1.5 T reference scanner)

Measurement	Bias	Lower LoA	Upper LoA	SD	ICC	RDC	CoV (%)
Psoas muscle CSA (cm ²)	− 0.3	− 5.1	4.5	2.5	0.97	4.6	5.3
Psoas muscle Index (cm ² /m ²)	− 0.085	− 1.7	1.5	0.83	0.95	1.6	5.3
Skeletal muscle CSA (cm ²)	1.9	− 4.9	8.7	3.5	1.00	7.5	2.3
Skeletal muscle Index (cm ² /m ²)	0.71	− 1.8	3.2	1.3	0.99	2.8	2.4
SAT CSA (cm ²)	− 7.3	− 30	15	11	0.99	26	7.3
SAT Index (cm ² /m ²)	− 2.3	− 9.6	5.0	3.7	0.99	8.4	7.5
VAT CSA (cm ²)	8.6	− 31	49	20	0.92	42	10
VAT Index (cm ² /m ²)	2.4	− 8.8	14	5.7	0.90	12	10
VAT/SAT Ratio	0.076	− 0.16	0.31	0.12	0.90	0.27	15

CoV coefficient of variation (%), CSA cross-sectional area, ICC intraclass coefficient, LoA limits of agreement, RC repeatability coefficient, SAT subcutaneous adipose tissue, SD standard deviation, VAT visceral adipose tissue

3.5 Inter- and intra-analyst variability

Both inter- and intra-analyst variability measures demonstrated excellent agreement for all body composition measurements, with ICC values ranging from 0.98 to 1.0, and low CoV values (0.5% to 5.0%). Detailed performance metrics are reported in Supplementary Table S5 for inter-analyst variability and in Supplementary Table S6 for intra-analyst variability for all three analysts.

3.6 Inter-reader (analysts vs. radiologists) variability

Each trained analyst using the study software demonstrated performance equivalent to the gold standard, as defined by the median results of three expert radiologists using the device. Pearson correlation coefficients indicated near-perfect correlation across all measurements, scanner field strengths, and manufacturers (*r* values: 0.97 to 1, *p* < 0.001). The ICC values were excellent, ranging from 0.94 to 1 (Supplementary Table S7).

3.7 Inter-device variability

The performance of the study software, as assessed by three radiologists, was found to be largely equivalent to that of the radiologist using the reference software for the tested body composition measurements: VAT CSA, SAT CSA, psoas muscle CSA, skeletal muscle CSA (Supplementary Table S8). The reliability was generally good to excellent (ICC 0.84 to 1.0), except for skeletal muscle CSA at Siemens 1.5 T and Philips 3 T, and psoas muscle CSA at GE 3 T, where the reliability was moderate (ICC 0.60, 0.67, 0.65, respectively). Upon further examination, measurement discrepancies were identified in 4 (out of 36) cases. The differences were explored in detail through separate interviews with each radiologist, who independently scored the segmentation masks generated by the study software and the regulated reference software (the results are reported in Supplementary Table S9). The consensus among the radiologists was that the study software provided measurements that were either comparable to or superior to those from the reference software.

4 Discussion

In this study, we have demonstrated that single-slice MRI body composition measurements at the L3 vertebra level provide consistent and reliable results across a diverse cohort, including both lean and obese individuals. All measurements showed high scan-rescan repeatability and cross-scanner reproducibility, with minimal inter/intra-observer and inter-device variability.

Single-slice measurements at the L3 vertebra have been shown to correlate strongly with volumetric assessments of muscle and adipose tissue, offering reliable insights comparable to those derived from full volumetric techniques [38, 39, 51]. While previous technical evaluations have focused on volumetric assessments [35, 52, 53] or single-slice assessments

of paraspinal muscles [54, 55], our study provides a comprehensive investigation into the technical validity of single-slice body composition measurements of both muscle and adipose tissue.

The scan-rescan repeatability defines the sensitivity to change in longitudinal studies, where consistent measurements enable accurate monitoring of disease progression and treatment effects. We assessed scan-rescan repeatability using the coefficient of variation (CoV) and repeatability coefficient (RC) [56]. The CoV standardizes the spread of differences between repeated measurements relative to the mean. This approach facilitates comparison of test–retest repeatability across different body composition compartments and measurement units, providing a robust evaluation of the method's reliability for clinical applications. Our results demonstrate low CoV for scan-rescan repeatability across all single-slice measurements, (1.5% to 7.9%). These results align with previous studies, which reported comparable CoV values (1% to 13%) for volumetric assessments of adipose tissue, including SAT, VAT, and total adipose tissue, with higher variability noted for manually segmented metrics [52, 57–60]. The CoV values for scan-rescan repeatability were consistently higher than those for intra- and inter-analyst variability for skeletal muscle CSA, SAT CSA, and VAT CSA, suggesting that variations in the images themselves were the primary contributors to variance. Conversely, for psoas muscle CSA, the CoV for scan-rescan repeatability was consistently lower than the CoV for intra- and inter-analyst assessments. This suggests that analyst-related factors contribute more to the variance observed for this smaller structure, possibly due to its susceptibility to minor discrepancies in measurement techniques and interpretation.

The repeatability coefficient (RC), representing the smallest detectable difference with 95% confidence, provides context for the clinical applicability of our measurements. Our RC values (15 cm² for VAT and 12 cm² for SAT) are much lower than the changes reported in a recent meta-analysis of weight-loss drug interventions (32 cm² for VAT and 43 cm² for SAT) [20], indicating sufficient sensitivity for detecting clinically relevant changes, particularly useful in populations with elevated baseline adiposity.

Reproducibility across different scanners is crucial for multi-center studies and standardized clinical practice. Our study found higher cross-scanner reproducibility CoV for SAT (7.3%) and VAT (10%) CSA compared to previous volumetric measurements (3.4% and 4.4%, respectively) [39]. This difference may reflect the greater susceptibility of single-slice measurements to inter-scanner variations or could be influenced by subject-related factors.

All body composition measurements performed here exhibited high consistency in inter- and intra-analyst variability assessments, as well as inter-reader (analysts vs. radiologists) assessments, as tested across various scanner field strengths and manufacturers. The inter- and intra-observer agreement was excellent (ICC 0.98–1.0) and similar to the previously reported values for skeletal and paraspinal muscle CSA (ICC 0.95–1.0) [54, 61], and VAT volume in individuals with BMI values up to 48.6 kg/m² (ICC 0.99) [52]. Additionally, the intra-observer variability was comparable to the inter-observer variability, suggesting that the method is robust against individual observer differences, allowing for interchangeable use of operators across studies.

The single-slice MRI method offers an efficient alternative to volumetric techniques, with shorter acquisition (< 1 min) and processing times (~ 5 min). Its consistent scan-rescan reliability and cross-scanner reproducibility could potentially reduce patient numbers and shorten clinical trial durations, leading to cost saving. Rapid quantitative assessments that can be performed by technicians may facilitate structured reporting and consistent clinical decision-making. Additionally, in this study, multiple scans were successfully acquired for each participant across all sessions without any acquisition failures or patient exclusions. This demonstrates a level of reliability that is typically not achieved with full volumetric approaches.

There were several limitations in this study. The analyses relied on manual segmentation techniques, which can be time-consuming. Automated or semi-automated approaches could potentially improve reliability and efficiency, particularly for complex measurements, such as VAT, where the likelihood of measurement error is higher due to intricate anatomy and variability of this tissue. The prospective cohort size for the repeatability and reproducibility analysis was relatively small, though the sample size was calculated to be sufficient based on Bland–Altman estimates. The prospective assessment included participants with a BMI range of 18–31 kg/m², covering normal weight, overweight, and mildly obese individuals, but did not include those with severe obesity (BMI > 35 kg/m²). In contrast, our retrospective assessment spanned a broader BMI range, from 16.3 to 44.4 kg/m². Further, recent research has confirmed that the reliability of total adipose tissue measurements, for example, is maintained in obese patients [52]. In addition, while the primary objective of this study was to assess the technical validity of single-slice measurements of body composition, which provides a foundation for broader applications, evaluating their utility for measuring change in longitudinal studies was beyond the scope of this work.

In conclusion, this study establishes the technical validity of the MRI-based single-slice method at the L3 vertebral level for quantifying adipose and muscle tissue. The method demonstrates accuracy, repeatability, and reproducibility

across diverse body types, multiple MRI scanner field strengths, and manufacturers, with consistent results between trained analysts and experienced radiologists. While further validation for assessing change in longitudinal applications is warranted, this method paves the way for broader implementation in clinical practice and research, contributing to the standardization of body composition assessments.

Author contributions MDR, MP, CEH, LFC, HTB, MN and RS contributed to the study conception and design. Material preparation, data collection and analysis were performed by LN, TD, CEH, LFP, MN, RS, MP, GRS, and AM. The first draft of the manuscript was written by MN. Review and editing were performed by MN, MDR, HTB, MP, CEH, LN, and TD. All authors read and approved the final manuscript.

Funding The study was funded by Perspectum Ltd.

Data availability Data supporting the findings of this study are included within the manuscript and supplementary information. Additional data may be made available upon reasonable request to qualified researchers, subject to legal and regulatory requirements.

Declarations

Competing interests MN, LN, CEH, TD, LFC, RS, HTB, and MDR are employees at Perspectum Ltd, the company that developed VitruvianScan. HTB and MDR are also shareholders at Perspectum Ltd. MP is a consultant for and a shareholder at Perspectum.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

1. Boutari C, Mantzoros CS. A 2022 update on the epidemiology of obesity and a call to action: as its twin COVID-19 pandemic appears to be receding, the obesity and dysmetabolism pandemic continues to rage on. *Metabolism*. 2022;133: 155217.
2. Cawley J, Biener A, Meyerhoefer C, et al. Direct medical costs of obesity in the United States and the most populous states. *J Manag Care Spec Pharm*. 2021. <https://doi.org/10.18553/jmcp.2021.20410>.
3. Petermann-Rocha F, Balntzi V, Gray SR, et al. Global prevalence of sarcopenia and severe sarcopenia: a systematic review and meta-analysis. *J Cachexia Sarcopenia Muscle*. 2022;13(1):86–99.
4. Dennison EM, Sayer AA, Cooper C. Epidemiology of sarcopenia and insight into possible therapeutic targets. *Nat Rev Rheumatol*. 2017;13(6):340–7.
5. Wormser D, Kaptoge S, et al. Separate and combined associations of body-mass index and abdominal adiposity with cardiovascular disease: collaborative analysis of 58 prospective studies. *Lancet*. 2011;377(9771):1085–95.
6. Kovesdy CP, Furth SL, Zoccali C. Obesity and kidney disease. *Can J Kidney Health Dis*. 2017;4:2054358117698669.
7. Nawaz S, Chinnadurai R, Al-Chalabi S, et al. Obesity and chronic kidney disease: a current review. *Obes Sci Pract*. 2022;9(2):61–74.
8. Singh GM, Danaei G, Farzadfar F, et al. The age-specific quantitative effects of metabolic risk factors on cardiovascular diseases and diabetes: a pooled analysis. *PLoS ONE*. 2013;8(7): e65174.
9. Karlas T, Wiegand J, Berg T. Gastrointestinal complications of obesity: non-alcoholic fatty liver disease (NAFLD) and its sequelae. *Best Pract Res Clin Endocrinol Metab*. 2013;27(2):195–208.
10. Huttasch M, Roden M, Kahl S. Obesity and MASLD: is weight loss the (only) key to treat metabolic liver disease? *Metabolism*. 2024;157: 155937.
11. Pati S, Irfan W, Jameel A, Ahmed S, Shahid RK. Obesity and cancer: a current overview of epidemiology, pathogenesis, outcomes, and management. *Cancers (Basel)*. 2023;15(2):485.
12. Rathmell JC. Obesity, immunity, and cancer. *N Engl J Med*. 2021;384(12):1160–2.
13. Chen H, Huang X, Dong M, Wen S, Zhou L, Yuan X. The association between sarcopenia and diabetes: from pathophysiology mechanism to therapeutic strategy. *Diabetes Metab Syndr Obes*. 2023;16:1541–54.
14. Mesinovic J, Zengin A, Courten BD, Ebeling PR, Scott D. Sarcopenia and type 2 diabetes mellitus: a bidirectional relationship. *Diabet Metabolic Syndrome Obesi TargetsTherapy*. 2019;12:1057.
15. Cruz-Jentoft AJ, Sayer AA. Sarcopenia. *The Lancet*. 2019;393(10191):2636–46.
16. Park B, Bhat S, Xia W, et al. Consensus-defined sarcopenia predicts adverse outcomes after elective abdominal surgery: meta-analysis. *BJS Open*. 2023;7(4):065.
17. Thomas EL, Frost G, Taylor-Robinson SD, Bell JD. Excess body fat in obese and normal-weight subjects. *Nutr Res Rev*. 2012;25(1):150–61.
18. Prentice AM, Jebb SA. Beyond body mass index. *Obes Rev*. 2001;2(3):141–7.

19. Blundell JE, Dulloo AG, Salvador J, Frühbeck G. Beyond BMI - phenotyping the obesities. *Obes Facts*. 2014;7(5):322–8.
20. Xu J, Wan CS, Ktoris K, Reijnierse EM, Maier AB. Sarcopenia is associated with mortality in adults: a systematic review and meta-analysis. *Gerontology*. 2021;68(4):361–76.
21. Beaudart C, Zaaria M, Pasleau F, Reginster J-Y, Bruyère O. Health outcomes of sarcopenia: a systematic review and meta-analysis. *PLoS ONE*. 2017;12(1): e0169548.
22. Malmstrom TK, Miller DK, Simonsick EM, Ferrucci L, Morley JE. SARC-F: a symptom score to predict persons with sarcopenia at risk for poor functional outcomes. *J Cachexia Sarcopenia Muscle*. 2016;7(1):28–36.
23. Merlotti C, Ceriani V, Morabito A, Pontiroli AE. Subcutaneous fat loss is greater than visceral fat loss with diet and exercise, weight-loss promoting drugs and bariatric surgery: a critical review and meta-analysis. *Int J Obes*. 2017;41(5):672–82.
24. Meyer-Gerspach AC, Peterli R, Moor M, et al. Quantification of liver, subcutaneous, and visceral adipose tissues by MRI Before and after bariatric surgery. *OBES SURG*. 2019;29(9):2795–805.
25. Ross R, Rissanen J, Pedwell H, Clifford J, Shragge P. Influence of diet and exercise on skeletal muscle and visceral adipose tissue in men. *J Appl Physiol*. 1996;81(6):2445–55.
26. Hua X, Deng J-P, Long Z-Q, et al. Prognostic significance of the skeletal muscle index and an inflammation biomarker in patients with breast cancer who underwent postoperative adjuvant radiotherapy. *Curr Probl Cancer*. 2020;44(2): 100513.
27. Prado CM, Lieffers JR, McCargar LJ, et al. Prevalence and clinical implications of sarcopenic obesity in patients with solid tumours of the respiratory and gastrointestinal tracts: a population-based study. *Lancet Oncol*. 2008;9(7):629–35.
28. Donato B, Almeida R, Raimundo M, et al. Myosteatosis: an underrecognized risk factor for mortality in non-dialysis chronic kidney disease patients. *J Nephrol*. 2024. <https://doi.org/10.1007/s40620-024-02042-2>.
29. Heymsfield SB, Smith B, Robson MD. Measurement of total adiposity, regional fat depots, and ectopic fat. Boca Raton: CRC Press; 2023.
30. Nordez A, Jolivet E, Südhoff I, Bonneau D, de Guise JA, Skalli W. Comparison of methods to assess quadriceps muscle volume using magnetic resonance imaging. *J Magn Reson Imaging*. 2009;30(5):1116–23.
31. Selberg O, Burchert W, Graubner G, Wenner C, Ehrenheim C, Müller MJ. Determination of anatomical skeletal muscle mass by whole body nuclear magnetic resonance. *Basic Life Sci*. 1993;60:95–7.
32. West J, Dahlqvist Leinhard O, Romu T, et al. Feasibility of MR-based body composition analysis in large scale population studies. *PLoS ONE*. 2016;11(9): e0163332.
33. Newman D, Kelly-Morland C, Leinhard OD, et al. Test-retest reliability of rapid whole body and compartmental fat volume quantification on a widebore 3T MR system in normal-weight, overweight, and obese subjects. *J Magn Reson Imaging*. 2016;44(6):1464–73.
34. Borga M, West J, Bell JD, et al. Advanced body composition assessment: from body mass index to body composition profiling. *J Investig Med*. 2018;66(5):1–9.
35. Borga M, Ahlgren A, Romu T, Widholm P, Dahlqvist Leinhard O, West J. Reproducibility and repeatability of MRI-based body composition analysis. *Magn Reson Med*. 2020;84(6):3146–56.
36. Mathews JD, Forsythe AV, Brady Z, et al. Cancer risk in 680 000 people exposed to computed tomography scans in childhood or adolescence: data linkage study of 11 million Australians. *BMJ*. 2013;346: f2360.
37. Power SP, Moloney F, Twomey M, James K, O'Connor OJ, Maher MM. Computed tomography and patient risk: facts, perceptions and uncertainties. *World J Radiol*. 2016;8(12):902–15.
38. Maislin G, Ahmed MM, Gooneratne N, et al. Single Slice vs. volumetric MR assessment of visceral adipose tissue - reliability and validity among the overweight and obese. *Obesity*. 2012;20(10):2124–32.
39. Schweitzer L, Geisler C, Pourhassan M, et al. What is the best reference site for a single MRI slice to assess whole-body skeletal muscle and adipose tissue volumes in healthy adults?1. *Am J Clin Nutr*. 2015;102(1):58–65.
40. Paris MT, Tandon P, Heyland DK, et al. Automated body composition analysis of clinically acquired computed tomography scans using neural networks. *Clin Nutr*. 2020;39(10):3049–55.
41. Borrelli P, Kaboteh R, Enqvist O, et al. Artificial intelligence-aided CT segmentation for body composition analysis: a validation study. *Euro Radiol Exp*. 2021;5(1):11.
42. Addeman BT, Kutty S, Perkins TG, et al. Validation of volumetric and single-slice MRI adipose analysis using a novel fully automated segmentation method. *J Magn Reson Imaging*. 2015;41(1):233–41.
43. Santini F, Pansini M, Deligianni X, Caligiuri ME, Oei EHG. ESR Essentials: advanced MR safety in vulnerable patients—practice recommendations by the European society for magnetic resonance in medicine and biology. *Eur Radiol*. 2024. <https://doi.org/10.1007/s00330-024-11055-1>.
44. Yu H, McKenzie CA, Shimakawa A, et al. Multiecho reconstruction for simultaneous water-fat decomposition and T2* estimation. *J Magn Reson Imaging*. 2007;26(4):1153–61.
45. Euser AM, Dekker FW, le Cessie S. A practical approach to Bland-Altman plots and variation coefficients for log transformed variables. *J Clin Epidemiol*. 2008;61(10):978–82.
46. Bland JM, Altman DG. Statistical methods for assessing agreement between two methods of clinical measurement. *Lancet*. 1986;1(8476):307–10.
47. Bland JM, Altman DG. Applying the right statistics: analyses of measurement studies. *Ultrasound Obstet Gynecol*. 2003;22(1):85–93.
48. Raunig DL, McShane LM, Pennello G, et al. Quantitative imaging biomarkers: a review of statistical methods for technical performance assessment. *Stat Methods Med Res*. 2015;24(1):27–67.
49. Koo TK, Li MY. A Guideline of selecting and reporting intraclass correlation coefficients for reliability research. *J Chiropr Med*. 2016;15(2):155–63.
50. Lu M-J, Zhong W-H, Liu Y-X, Miao H-Z, Li Y-C, Ji M-H. Sample size for assessing agreement between two methods of measurement by bland-Altman method. *Int J Biostat*. 2016. <https://doi.org/10.1515/ijb-2015-0039>.
51. Schaidinn A, Linder N, Garnov N, et al. Predictive accuracy of single- and multi-slice MRI for the estimation of total visceral adipose tissue in overweight to severely obese patients. *NMR Biomed*. 2015;28(5):583–90.
52. Newman D, Kelly-Morland C, Leinhard OD, et al. Test-retest reliability of rapid whole body and compartmental fat volume quantification on a widebore 3T MR system in normal-weight, overweight, and obese subjects. *J Magn Reson Imaging*. 2016;44(6):1464–73.

53. Middleton MS, Haufe W, Hooker J, et al. Quantifying abdominal adipose tissue and thigh muscle volume and hepatic proton density fat fraction: repeatability and accuracy of an MR imaging-based. *Semauto Anal Method Radiol*. 2017;283(2):438–49.
54. Ropponen A, Videman T, Battié MC. The reliability of paraspinal muscles composition measurements using routine spine MRI and their association with back function. *Man Ther*. 2008;13(4):349–56.
55. Hu Z-J, He J, Zhao F-D, Fang X-Q, Zhou L-N, Fan S-W. An assessment of the intra- and inter-reliability of the lumbar paraspinal muscle parameters using CT scan and magnetic resonance imaging. *Spine*. 2011;36(13):E868–874.
56. Obuchowski NA. Interpreting change in quantitative imaging biomarkers. *Acad Radiol*. 2018;25(3):372–9.
57. Gerard EL, Snow RC, Kennedy DN, et al. Overall body fat and regional fat distribution in young women: quantification with MR imaging. *AJR Am J Roentgenol*. 1991;157(1):99–104.
58. Silver HJ, Niswender KD, Kullberg J, et al. Comparison of gross body fat-water magnetic resonance imaging at 3 Tesla to dual-energy X-ray absorptiometry in obese women. *Obesity (Silver Spring)*. 2013;21(4):765–74.
59. Kullberg J, Johansson L, Ahlström H, et al. Automated assessment of whole-body adipose tissue depots from continuously moving bed MRI: a feasibility study. *J Magn Reson Imaging*. 2009;30(1):185–93.
60. Bonekamp S, Ghosh P, Crawford S, et al. Quantitative comparison and evaluation of software packages for assessment of abdominal adipose tissue distribution by magnetic resonance imaging. *Int J Obes*. 2008;32(1):100–11.
61. Kiefer LS, Fabian J, Lorbeer R, et al. Inter- and intra-observer variability of an anatomical landmark-based, manual segmentation method by MRI for the assessment of skeletal muscle fat content and area in subjects from the general population. *Br J Radiol*. 2018;91(1089):20180019.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.